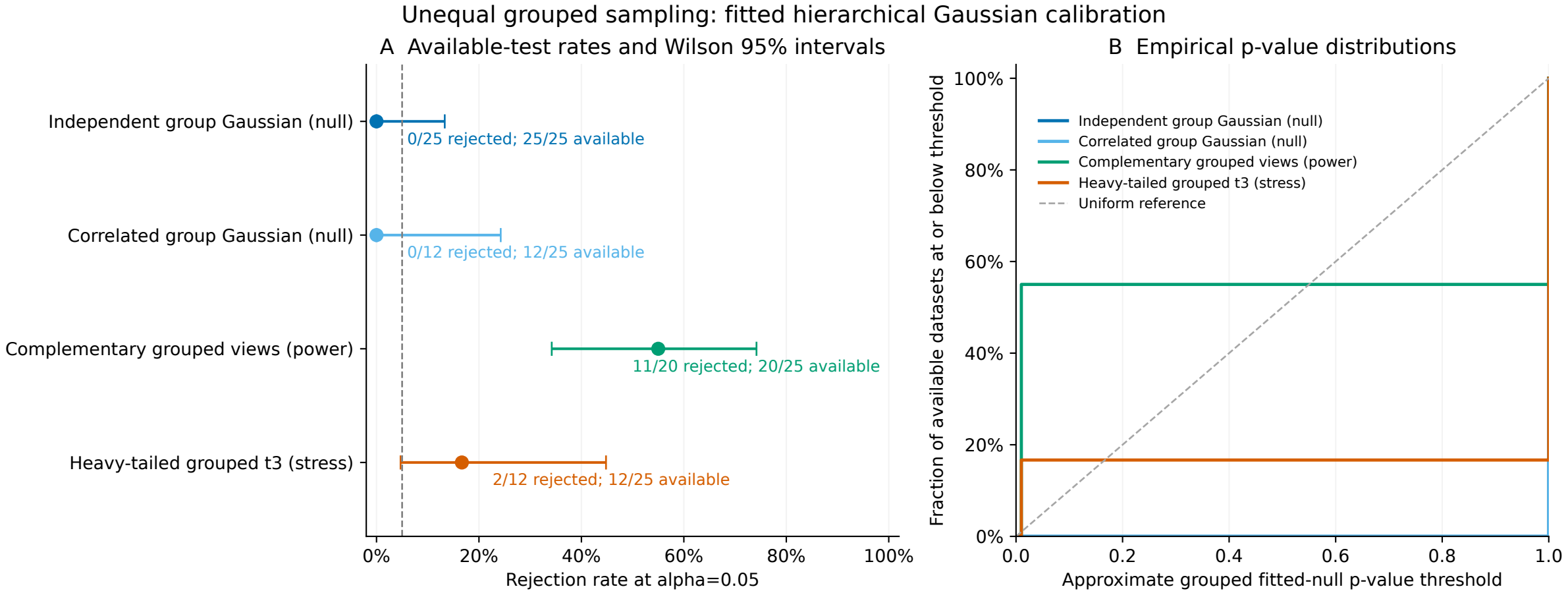
Unequal grouped sampling: fitted hierarchical Gaussian calibration
A Available-test rates and Wilson 95% intervals
Independent group Gaussian (null)
0/25 rejected; 25/25 available
Correlated group Gaussian (null)
0/12 rejected; 12/25 available
Complementary grouped views (power)
11/20 rejected; 20/25 available
Heavy-tailed grouped t3 (stress)
2/12 rejected; 12/25 available
0%
20%
40%
60%
80%
100%
Rejection rate at alpha=0.05
B Empirical p-value distributions
Fraction of available datasets at or below threshold
Independent group Gaussian (null)
Correlated group Gaussian (null)
Complementary grouped views (power)
Heavy-tailed grouped t3 (stress)
Uniform reference
0.0
0.2
0.4
0.6
0.8
1.0
Approximate grouped fitted-null p-value threshold